

Survival Analysis - Assignment 2

2026-03-12

Q1 Frailty model

In a frailty model, the hazard rate at time t and conditional on a frailty Z is

$$\alpha(t|Z) = Z \cdot \alpha(t).$$

Given Z , the survival function to time t is

$$S(t|Z) = \exp\left(-Z \int_0^t \alpha(u) du\right) = \exp(-ZA(t)).$$

Introducing the Laplace transform and the population function is found by integrating over the distribution of Z

$$S(t) = \int_0^t f_Z(z) e^{-zA(t)} dz = E_Z(e^{-zA(t)}) = \mathcal{L}(A(t)).$$

The population hazard rate is

$$\mu(t) = -\frac{d}{dt} \log(\mathcal{L}(A(t))) = -\frac{\frac{d}{dt} \mathcal{L}(A(t))}{\mathcal{L}(A(t))} = \alpha(t) \frac{-\mathcal{L}'(A(t))}{\mathcal{L}(A(t))}$$

Naming the coefficient as $G(A(t))$, $G(A(t)) = \frac{-\mathcal{L}'(A(t))}{\mathcal{L}(A(t))}$. Taking the first derivative of $G(A(t))$, we obtain

$$G'(A(t)) = \frac{-\mathcal{L}''(A(t))\mathcal{L}(A(t)) + (\mathcal{L}'(A(t)))^2}{[\mathcal{L}(A(t))]^2}.$$

With $\mathcal{L}(A(t)) = E(e^{-ZA(t)})$, $\mathcal{L}'(A(t)) = -E(Ze^{-ZA(t)})$, $\mathcal{L}''(A(t)) = E(Z^2 e^{-ZA(t)})$, the numerator of $G'(A(t))$ is then

$$-\left(E(Z^2 e^{-ZA(t)})\right) \cdot E(e^{-ZA(t)}) + \left(-E(Ze^{-ZA(t)})\right)^2$$

According to the Cauchy-Schwarz Inequality $[E(XY)]^2 \leq E(X^2) \cdot E(Y^2)$, the numerator is negative, indicating that the coefficient $G(A(t))$ is decreasing over t .

When $t = 0$, $A(0) = 0$, $\mathcal{L}(A(0)) = 1$, $\mathcal{L}'(A(0)) = E(Z) = 1$,

$$G(A(0)) = \frac{-\mathcal{L}'(A(0))}{\mathcal{L}(A(0))} = 1.$$

Thus when $t \geq 0$,

$$G(A(t)) \leq G(A(0)) = 1$$

$$\mu(t) = \alpha(t) \cdot G(A(t)) \leq \alpha(t).$$

Therefore, we reach the conclusion that the population hazard is less than or equal to $\alpha(t)$.

Q2 Poisson likelihood

First, we prove that this model can be fitted using Poisson regression in theory. The baseline hazard is $h_0(t) = 2\lambda t$, thus

$$h(t|x) = 2\lambda t \cdot \exp(\beta^T x),$$

$$H(t|x) = \int_0^t h(u|x) du = \lambda t^2 \cdot \exp(\beta^T x).$$

Assuming that $i \in \{1, \dots, n\}$, for right-censored data, the likelihood L is

$$L(\theta) = \prod_{i=1}^n h(t_i|x_i)^{\delta_i} \cdot S(t_i|x_i)$$

The full log-likelihood is then

$$\begin{aligned} l(\theta) &= \sum_{i=1}^n \{ \delta_i \log(h(t_i|x_i)) - H(t_i|x_i) \} \\ &= \sum_{i=1}^n \{ \delta_i \log(H(t_i|x_i)) - H(t_i|x_i) \} + \sum_{i=1}^n \delta_i \log\left(\frac{h(t_i|x_i)}{H(t_i|x_i)}\right) \end{aligned}$$

The first term follows the structure of a Poisson distribution and is a Poisson kernel. Therefore, δ_i can be viewed as a Poisson variable, and $H(t_i|x_i)$ can be regarded as the Poisson mean.

$$E(\delta_i) = H(t_i|x_i)$$

$$\begin{aligned} \log(E(\delta_i)) &= \log(H(t_i|x_i)) \\ &= \log(\lambda t_i^2 \exp(\beta^T x_i)) \\ &= \log \lambda + \beta^T x_i + 2\log(t_i) \end{aligned}$$

The result shows that the offset of this Poisson regression model is $2\log(t_i)$.

Next, we estimate β, λ with simulated data. Since $S \in [0, 1]$, we assume S follows the standard uniform distribution, and we obtain t :

$$S = \exp(-H(t)) = \exp(-\lambda t^2 \cdot \exp(\beta^T x))$$

$$\log U = -\lambda t^2 \cdot \exp(\beta^T x)$$

$$t = \sqrt{\frac{-\log U}{\lambda \exp(\beta^T x)}}$$

The simulation process is hereby:

```
set.seed(1125)

# set variable
n <- 1000
lambda_true <- 0.5
beta_true <- 1.2
x <- rnorm(n)
u <- runif(n)

# construct t, delta
t_event <- sqrt(-log(u)/(lambda_true*exp(beta_true*x)))
censoring_time <- 1.5
t_obs <- pmin(t_event, censoring_time)
delta <- as.numeric(t_event <= censoring_time)

# build poisson model
X <- cbind(1, x)
fit <- glm.fit(x = X, y = delta,
               offset = 2*log(t_obs),
               family = poisson())

# extract estimation value
beta_est <- coef(fit)["x"]
lambda_est <- exp(coef(fit)[1])

cat("\n--- Estimation Result ---",
    "\nBeta Estimated:", round(beta_est, 4), " (True:", beta_true,")",
    "\nLambda Estimated:", round(lambda_est, 4), " (True:", lambda_true,
    ")")

##
## --- Estimation Result ---
## Beta Estimated: 1.193 (True: 1.2 )
## Lambda Estimated: 0.501 (True: 0.5 )

# obtain std_error
W <- fit$weights
info_matrix <- t(X) %*% (W * X)
vcov_matrix <- solve(info_matrix)
std_er <- sqrt(diag(vcov_matrix))
beta_se <- std_er[2]
```

```

# Delta method
lambda_se <- lambda_est * std_er[1] # g'(theta) * SE(theta)

# print results
cat("\n")

cat("SE of Beta:", beta_se, "\n")

## SE of Beta: 0.0452556

cat("SE of Lambda:", lambda_se, "\n")

## SE of Lambda: 0.02241052

```

Q3 Weibull Regression

To prove that the Weibull distribution is the only distribution which is both Proportional Hazards (PH) and Accelerated Failure Time (AFT), is to show that, if a survival function $S_0(t)$ satisfies the identity $[S_0(t)]^\psi = S_0(t/\phi)$ for all $\psi, \phi > 0$, then $S_0(t)$ must take the form of a Weibull survival function. We derive the proof as below:

$$[S_0(t)]^\psi = S_0(t/\phi)$$

Taking the logarithm of both sides and with the relationship between S and $H(H_0(t) = -\ln S_0(t))$, we obtain

$$\begin{aligned}\psi \cdot \ln(S_0(t)) &= \ln(S_0(t/\phi)) \\ \psi H_0(t) &= H_0(t/\phi)\end{aligned}$$

Setting $t = 1$ and substituting back,

$$\begin{aligned}\psi &= \frac{H_0(1/\phi)}{H_0(1)} \\ \frac{H_0(1/\phi)}{H_0(1)} \cdot H_0(t) &= H_0(t/\phi)\end{aligned}$$

Dividing both sides by $H_0(1)$ and define $M(k) = \frac{H_0(k)}{H_0(1)}$,

$$\begin{aligned}M(1/\phi) \cdot M(t) &= M(t/\phi) \\ \frac{dM(1/\phi)}{d\phi} \cdot M(t) &= \frac{dM(t/\phi)}{d\phi} \\ -\frac{1}{\phi^2} M'(1/\phi) \cdot M(t) &= -\frac{t}{\phi^2} M'(t/\phi)\end{aligned}$$

Setting $\phi = 1$ and solve the ODE

$$M'(1) \cdot M(t) = t \cdot M'(t)$$

$$\frac{M'(t)}{M(t)} = \frac{M'(1)}{t}$$

$$\ln M(t) = M'(1) \ln t + C$$

When $t = 1$, we got $M'(1) \cdot M(1) = 1 \cdot M'(1)$, i.e. $M(1) = 1$. We also have $\ln M(1) = M'(1) \ln 1 + C$, thus $C = 0$. Therefore, $M(t)$ has the solution

$$M(t) = t^{G'(1)} = t^\gamma$$

Transforming back to the H_0 , and denote $H_0(1) = \lambda$,

$$H_0(t) = \lambda \cdot t^\gamma$$

$$S_0(t) = \exp(-H_0(t)) = \exp(-\lambda \cdot t^\gamma)$$

Since $S_0(t)$ here follows the standard form of a Weibull distribution, the proof is done.

Q4 RMST calculations

To obtain RMST from a survPen model, we fit the model by specifying penalized restricted cubic splines for time and including treat as the main exposure variable. To calculate 95% confidence intervals, delta method is used and the derivation is:

$$\text{grad}_{\text{RMST}} = \frac{\partial \text{RMST}}{\partial \beta} = \frac{\partial \int S dt}{\partial \beta} = \int \frac{\partial \exp(-H)}{\partial \beta} dt = \int -S \frac{\partial H}{\partial \beta} dt = \int \text{grad}_S dt$$

$$\text{grad}_S = -S \cdot \frac{\partial H}{\partial \beta} = -S \cdot \text{grad}_H$$

$$\text{grad}_H = \frac{\partial H}{\partial \beta} = \frac{\partial \int h dt}{\partial \beta} = \int \frac{\partial \exp(\eta)}{\partial \beta} dt = \int \frac{\partial \exp(\beta X)}{\partial \beta} dt = \int h \cdot X dt$$

Thus,

$$\text{Var}(\text{RMST}) = (\text{grad}_{\text{RMST}})^T \cdot \text{Var}(\beta) \cdot (\text{grad}_{\text{RMST}})$$

hint: delta method; make a plot/table for difference

install.packages("survPen")

library("survPen")

data_bre <- collett::breast_rfs

head(data_bre)

```
##   id treat age men size grade nodes prog oest time status
## 1  1     0  70  2  21    2     3   48  66 1814      1
## 2  2     1  56  2  12    2     7   61  77 2018      1
```

```

## 3 3      1 58 2 35      2 9 52 271 712      1
## 4 4      1 59 2 17      2 4 60 29 1807      1
## 5 5      0 73 2 35      2 1 26 65 772      1
## 6 6      0 32 1 57      3 24 0 13 448      1

model_Pen <- survPen(~ treat + smf(time),
                     data = data_bre,
                     t1 = time,
                     event = status)

t_seq <- seq(0, 1826, length.out = 100)
dt <- t_seq[2] - t_seq[1]

# first non-treated then treated
pred_data <- data.frame(
  time = rep(t_seq, 2),
  treat = rep(c(0, 1), each = length(t_seq))
)

pred.pen <- predict(model_Pen, newdata = pred_data)
h_t <- pred.pen$haz
S_t <- pred.pen$surv

# point estimation
rmst_est0 <- sum(S_t[1:100]) * dt
rmst_est1 <- sum(S_t[101:200]) * dt
rmst_diff <- rmst_est1 - rmst_est0

# interval estimation
X_pen <- predict(model_Pen, newdata = pred_data, type = "lpmatrix")
Var_beta <- model_Pen$Vp
grad_H0 <- apply(X_pen[1:100,] * h_t[1:100] * dt, 2, cumsum)
grad_S0 <- -S_t * grad_H0

grad_H1 <- apply(X_pen[101:200,] * h_t[101:200] * dt, 2, cumsum)
grad_S1 <- -S_t * grad_H1

grad_RMST0 <- colSums(grad_S0)*dt
grad_RMST1 <- colSums(grad_S1)*dt

Var_est0 <- as.numeric(t(grad_RMST0) %*% Var_beta %*% grad_RMST0)
se_est0 <- sqrt(Var_est0)
Var_est1 <- as.numeric(t(grad_RMST1) %*% Var_beta %*% grad_RMST1)
se_est1 <- sqrt(Var_est1)

grad_diff <- grad_RMST1 - grad_RMST0
Var_diff <- as.numeric(t(grad_diff) %*% Var_beta %*% grad_diff)
se_diff <- sqrt(Var_diff)

```

```

results <- data.frame(
  Term = c("RMST0 (T0)", "RMST1 (T1)", "RMST Difference (T1 - T0)"),
  Estimate = c(rmst_est0, rmst_est1, rmst_diff),
  Std.Error = c(se_est0, se_est1, se_diff),
  Conf.Low = c(
    rmst_est0 - 1.96 * se_est0,
    rmst_est1 - 1.96 * se_est1,
    rmst_diff - 1.96 * se_diff
  ),
  Conf.High = c(
    rmst_est0 + 1.96 * se_est0,
    rmst_est1 + 1.96 * se_est1,
    rmst_diff + 1.96 * se_diff
  )
)

print(results)

```

	Term	Estimate	Std.Error	Conf.Low	Conf.High
## 1	RMST0 (T0)	1280.4260	31.58674	1218.5160	1342.3360
## 2	RMST1 (T1)	1422.2373	37.78931	1348.1702	1496.3043
## 3	RMST Difference (T1 - T0)	141.8112	49.60456	44.5863	239.0362

Q5 Likelihoods

From the setting of the question, we only know the months of registration and events, rather than the exact time. So we do the following assumption: Assume the month of registration is $M_R = 0$ and the month of event is $M_E = k$. Since the registration is distributed across a month, we further assume the exact registering time t_R is uniformly distributed, that is, $t_R \sim U(0,1)$.

Status{1,2}:

Similarly, t_E for status $\{1,2,3\}$ is $t_E \sim U(k, k+1)$. For status $\{1,2\}$, the survival time $T = t_E - t_R$. Here T is the difference of two uniformly distributed random variables, thus it can be expressed as a convolution process. With t_E, t_R setted, it can be considered as the exact time occasion, and the corresponding likelihood is

$$\begin{aligned}
 L &= \int_0^1 \int_k^{k+1} f(t_E - t_R) dt_E dt_R \\
 &= \int_0^1 F(k+1 - t_R) - F(k - t_R) dt_R \\
 &= \int_0^1 S(k - t_R) - S(k+1 - t_R) dt_R
 \end{aligned}$$

Status{3}:

For status {3}, we know that the Survival time $T > t_E - t_R$, it's the right-censored occasion. The corresponding likelihood is

$$\begin{aligned} L &= Pr(T > t_E - t_R) \\ &= \int_0^1 \int_k^{k+1} S(t_E - t_R) dt_E dt_R \end{aligned}$$

Status{4}:

As the event time here is assumed to be the end of the month, we got $t_E = k + 1$. It is also a right-censored occasion and the survival time is then $T > k + 1 - t_R$. The corresponding likelihood is

$$\begin{aligned} L &= Pr(T > k + 1 - t_R) \\ &= \int_0^1 S(k + 1 - t_R) dt_R \end{aligned}$$

Q6 Competing risks

a)

1. Difference between the cause-specific hazard ratio(csHR) and the subdistribution hazard ratio(sHR):

$$csHR = \frac{h_j(t|X = 1)}{h_j(t|X = 0)}$$

where

$$\begin{aligned} h_j(t|X = x) &= \lim_{\Delta t \rightarrow 0} \frac{P(t \leq T < t + \Delta t, C = j | T \geq t, X = x)}{\Delta t} \\ sHR &= \frac{\lambda_j(t|X = 1)}{\lambda_j(t|X = 0)} \end{aligned}$$

where

$$\lambda_j(t|X = x) = \lim_{\Delta t \rightarrow 0} \frac{P(t \leq T < t + \Delta t, C = j | T \geq t \cup (T < t, C \neq j), X = x)}{\Delta t}$$

Both ratios quantify the instantaneous rate of the event of interest (PCM), but they differ in the risk sets (the denominator of the hazard function). * For csHR, the risk set is people who are event-free. In this model, an individual is removed from the risk set once they experienced PCM or death, and the competing event is treated as censored. * For sHR, the risk set consists of all people who have not experienced the event of interest(PCM), that is, individuals who experienced the competing event are retained in the risk set.

2. Why do the estimated hazard ratios for age differ It is shown in the output that $csHR = 1.024 > 1$ and $cHR = 0.983 < 1$. This occurs because as age

increases, the rate of competing deaths in the elderly outpaces the rate of PCM progression, leading to a lower subdistribution hazard despite a higher cause-specific hazard.

3. What does each one tell us about the effect of age on PCM * $csHR = 1.024$ implies that every unit increase in age leads to a higher risk of progressing to PCM by around 2.4%. * $cHR = 0.983$ demonstrates that when people get older, the instantaneous risk decreases 1.7% per year because they have a higher probability of dying.

b)

```
library(mstate)

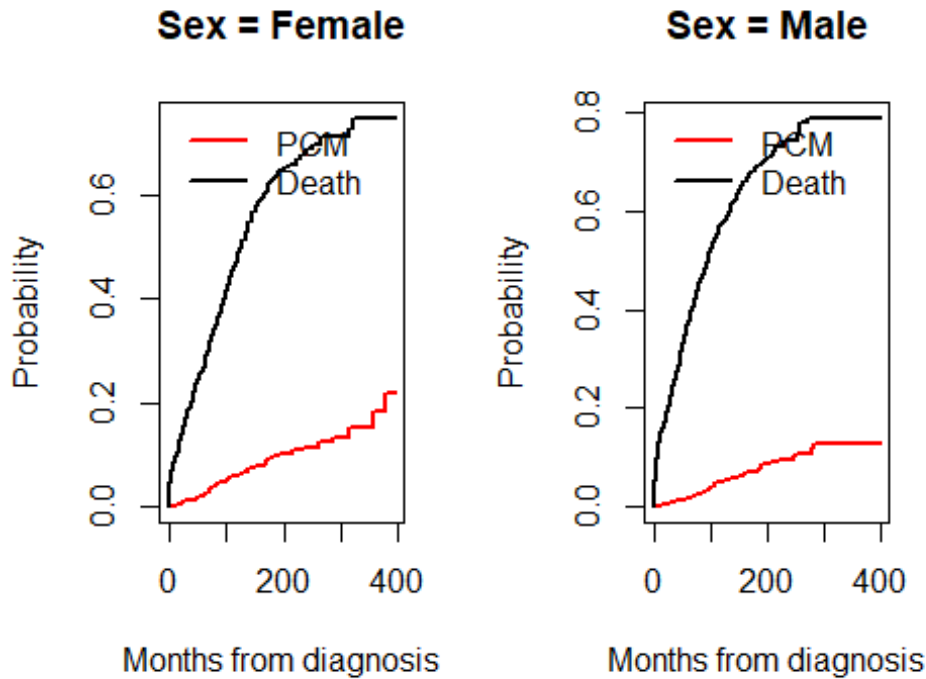
# Label event type
mgus2$event_type <- factor(with(mgus2, ifelse(pstat == 1, 1,
                                             ifelse(death == 1, 2, 0))),
                           levels = 0:2,
                           labels = c("censored", "PCM", "Death"))

# stratify by sex
data_f <- subset(mgus2, sex == "F")
data_m <- subset(mgus2, sex == "M")

# construct two Aalen-Johansen separately
ajfit_f <- survfit(Surv(futime, event_type) ~ 1, data = data_f)
ajfit_m <- survfit(Surv(futime, event_type) ~ 1, data = data_m)

# plot CIF
# max(data_f$futime)
par(mfrow = c(1, 2))
plot(ajfit_f, col = c("red", "black"), xmax = 400, lwd = 2,
     xlab = "Months from diagnosis", ylab = "Probability",
     main = "Sex = Female")
legend("topleft", c("PCM", "Death"), col = c("red", "black"), lty = 1,
      lwd = 2, bty = "n")

# max(data_m$futime)
plot(ajfit_m, col = c("red", "black"), xmax = 400, lwd = 2,
     xlab = "Months from diagnosis", ylab = "Probability",
     main = "Sex = Male")
legend("topleft", c("PCM", "Death"), col = c("red", "black"), lty = 1,
      lwd = 2, bty = "n")
```



Yes, male and female patients have different cumulative risks of PCM over time. Observing the plots, the risk of female patients shows a steady and consistent increase throughout the follow-up, and there is a sharper rise toward the end of the observation period, with the cumulative incidence eventually reaching approximately 0.22. While for male patients, the progression is more gradual compared to females. The curve appears to stabilize at around 0.15 starting from approximately the 300-month.

c)

- i) When death is present as a competing risk, Kaplan-Meier estimator still treat it as censored. It corresponds to the CIF as below

$$\begin{aligned}
 S_j(t) &= 1 - \int_0^t h_j(u) \exp(-H_j(u)) du \\
 &\leq 1 - \int_0^t h_j(u) \exp\left(-\sum_{j=1}^K H_j(u)\right) du = 1 - F_j(t),
 \end{aligned}$$

$$\begin{aligned}
F_{jKM}(t) &= 1 - S_{jKM}(t) \\
&= \int_0^t h_j(u) \exp(-H_j(u)) du \\
F_{jAJ}(t) &= 1 - S_j(t) \\
&= \int_0^t h_j(u) \exp\left(-\sum_{j=1}^K H_j(u)\right) du
\end{aligned}$$

As shown in the formulas, the “fake” KM-based CIF (F_{jKM}) integrates the hazard over a survival probability that only considers one cause, whereas the true CIF F_{jAJ} (which is also the AJ-estimator) integrates over the all-cause survival probability $S_{all}(t) = \exp(-\sum H_k(t))$. This leads to $F_{jKM}(t) \geq F_{jAJ}(t)$, proving that KM consistently overestimates the risk of the PCM when competing risks are present, while Aalen-Johansen estimator provides unbiased estimates.

ii) Plot of two estimators

```

# AJ-estimator
aj_fit <- survfit(Surv(futime, event_type) ~ 1, data = mgus2)

# KM-estimator
km_fit <- survfit(Surv(futime, pstat) ~ 1, data = mgus2)

# plot
plot(aj_fit[, "PCM"],
     conf.int = FALSE,
     col = "blue",
     lwd = 2,
     # xmax = 400,
     ylim = c(0, 1),
     xlab = "Months",
     ylab = "Cumulative Incidence of PCM",
     main = "Comparison of AJ and KM Estimators")

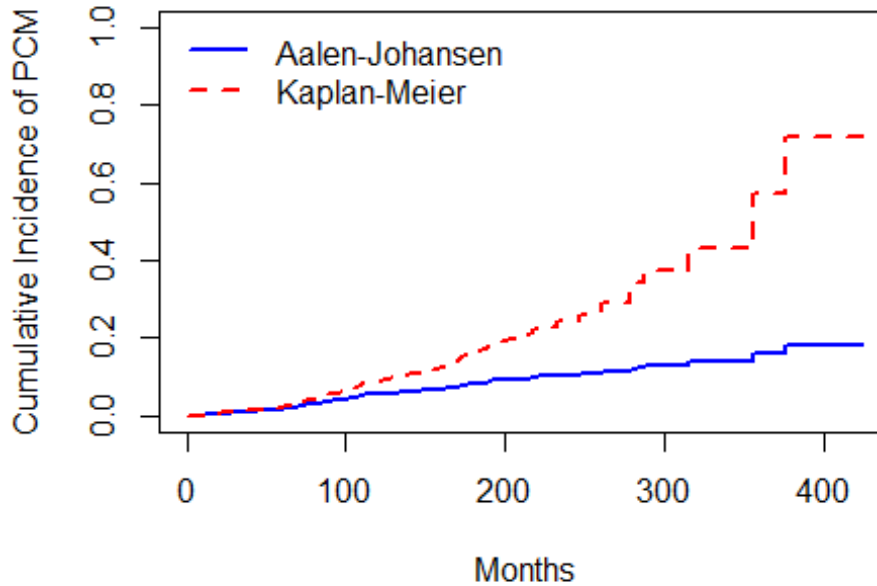
lines(km_fit,
      fun = "event",
      conf.int = FALSE,
      col = "red",
      lwd = 2,
      lty = 2)

legend("topleft",
      c("Aalen-Johansen", "Kaplan-Meier"),
      col = c("blue", "red"),
      lty = c(1, 2),

```

```
lwd = 2,
bty = "n")
```

Comparison of AJ and KM Estimators



As shown in the figure, the K-M estimator always yields a higher CIF of PCM comparing to the A-J estimator, demonstrating an overestimated bias.

Q7 Net survival

- (a) These two estimators differ mainly in how they weight individuals based on their expected mortality. Ederer II Estimator:

$$\hat{\Lambda}_E^*(t) = \int_0^t \frac{dN(u)}{Y(u)} - \int_0^t \frac{\sum_{i=1}^n Y_i(u) d\Lambda_{Pi}(u)}{\sum_{i=1}^n Y_i(u)}$$

In Ederer II estimator, the expected mortality rate $d\Lambda_{Pi}(u)$ appears only in the numerator of the subtrahend and is weighted by $Y_i(u)$, representing the individuals currently at risk. This arithmetic mean term reflects the average background risk of the remaining survivors at each time point. However, as individuals with higher background mortality are progressively removed from the risk set, the risk set shifts towards those mainly contains people with lower background risk. Consequently, Ederer II fails to provide an adequate correction. Pohar-Perme Estimator:

$$\widehat{\Lambda}_E(t) = \int_0^t \frac{dN^w(u)}{Y^w(u)} - \int_0^t \frac{\sum_{i=1}^n Y_i^w(u) d\Lambda_{Pi}(u)}{Y^w(u)}$$

$$N_i^w(t) = \frac{N_i(t)}{S_{Pi}(t)}, Y_i^w(t) = \frac{Y_i(t)}{S_{Pi}(t)}$$

In Pohar-Perme Estimator, the expected mortality rate is used through inverse-probability weighting (IPW): $1/S_{Pi}(u) = 1/\exp(-\int_0^u \lambda_{Pi}(s)ds)$. In this way, for individuals with a high background mortality, they have a huge weight too, effectively eliminating the influence of high mortality rate, thus leading to an unbiased estimation of net survival.

- (b) There exists a bias of Ederer II when background mortality is high, because it fails to properly account for informative censoring caused by high background mortality. Mathematically, the net hazard that Ederer II estimates is

$$\lambda_E^*(t) = \frac{\sum_i S_i^O(t) \lambda_{Ei}(t)}{\sum_i S_i^O(t)},$$

where $S_i^O(t) = S_i^E(t)S_i^P(t)$ is the observable net survival probability. While the true net hazard is

$$\lambda_E(t) = \frac{\sum_i S_i^E(t) \lambda_{Ei}(t)}{\sum_i S_i^E(t)}.$$

For individuals whose background mortality is high, they die faster from other causes and are removed from the risk earlier. High in λ also leads to expected survival (S_{Pi}) dropping rapidly, causing such patients to be down-weighted faster in the Ederer II calculation. This down-weighting them creates a selection bias that the risk set shifts towards those mainly contains people with lower background risk. Consequently, the Ederer II estimator underestimates the expected hazard, leading to an overestimation in net survival.

AI declaration:

In Q2, when I want to use “glm” function to build model, my R explode for no reason, so I use “glm.fit” function, and use AI to compute stand error since I can not get from “summary fit” here.

In Q4, I use AI to construct my answers into a structured data frame.